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Research Article

AI-RAN: The pathway to future wireless networks

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ABSTRACT

With the rapid advancement of Artificial Intelligence (AI), the Radio Access Network (RAN) is poised to undergo a transformative evolution toward the convergence of AI and RAN functionalities, referred to as the AI-RAN paradigm. AI-RAN integrates high-performance computing resources into RAN infrastructures, thereby enabling the execution of both AI and RAN workloads on the same infrastructure. As a result, it improves resource utilization, reduces energy consumption, and promotes swift AI-related responses. In this paper, we provide a comprehensive overview of AI-RAN, whereby we broadly categorize the discussion into three aspects: AI and RAN, AI for RAN, and AI on RAN. In particular, we begin with AI and RAN, which encompass the hardware architecture, software stack, as well as orchestration of computational and communication resources. We subsequently elaborate on the AI on RAN, examining various approaches to leveraging AI methods to enhance RAN performance. For the topic of AI on RAN, we conduct an in-depth investigation into the schemes that take RAN as a platform to facilitate AI services, where we review distributed learning for multi-cell and multi-vendor RANs, including federated and multi-agent reinforcement learning, highlighting issues of data heterogeneity, control-plane overhead, convergence under mobility, privacy, and adversarial robustness in the RAN ecosystems. We also demonstrate several use cases pertaining to the AI-RAN framework. We conclude by outlining key open issues and research directions.

1. Introduction

What will 6G be? Among the various anticipated features, ranging from integrated sensing and communication systems, terahertz communications, to digital twins and semantic communications, the (most widely) consensus is *network intelligence*. Indeed, the 6G vision weighs native intelligence as important as communication and computing, forging a paradigm shift from Artificial Intelligence (AI)-enabled system optimization to AI-native Radio Access Networks (RANs) where AI is integrated into the RAN architecture, air interface, and operational control [1,2]. Recent operator and standardization perspectives emphasize that, unlike 5G networks, where

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Machine Learning (ML) has been largely retrofitted, the 6G RAN is expected to be designed with intrinsic AI capabilities, including built-in data collection, model distribution, and closed-loop control across protocol layers. In parallel, the disaggregation and cloudification of RAN, which is even accelerated by the emergence of Open RAN (O-RAN) [3], are opening standardized control surfaces, e.g., the near-Real Time (RT) and/or non-RT RAN Intelligent Controller (RIC) and E2, A1, and O1 interfaces, that enable multi-vendor to test and implement AI applications [4]. Consequently, it triggers a convergence between AI and RAN, forming the AI-RAN revolution, as illustrated in Fig. 1.

However, making intelligence intrinsic to RAN poses new system-level challenges. Specifically, end-user data are inherently distributed, non-Independent and Identically Distributed (IID), and privacy-sensitive, motivating distributed learning paradigms (including federated, hierarchical, and multi-agent Reinforcement Learning (RL)) as the default for multi-cell inference and control [5–7]. Yet distributed learning in RAN must compensate for mobility-induced non-stationary, straggler effects, and control-plane overhead that can destabilize RT decisions. Moreover, AI-native RAN is inseparable from edge computing: the xApps/rApps, physical (PHY)-adjacent inference, and semantic tasks require cloud-edge-device co-training and low-latency deployment on heterogeneous accelerators [8,9]. Emerging AI-RAN platform efforts highlight both the feasibility and the urgency of co-locating RAN signal processing with AI workloads on shared edge or cloud infrastructure [10]. In addition, the emergence of task-oriented communications requires a fundamental departure from the air interference design: For many AI services, the objective is no longer bit-perfect reconstruction but task success, i.e., transmitting only the meaning or features required for downstream inference or control [11–15]. Early studies in task-oriented communication demonstrate substantial efficiency gains through joint source-channel-task design, yet also raise open questions regarding semantic fidelity metrics, robustness to domain shift, and interoperability with existing RAN layers. Finally, AI-RAN will not be deployed at scale without Service Management and Orchestration (SMO) that treats models, data, and compute as elementary resources. The O-RAN framework and ongoing 3rd Generation Partnership Project (3GPP) activities stress that standards will increasingly focus on AI lifecycle management—data exposure, model activation, performance monitoring, and safe fallback—rather than standardizing specific models.

Motivated by the requirements of the aforementioned new features, this paper provides a comprehensive introduction to AI-RAN in the foreseeable 6G era. Specifically, we target at covering: (1) an overview of the hardware architecture that supports a cohesive processing of AI and RAN workloads, (2) an in-depth survey of the AI applications that can be leveraged to enhance the RAN performance, (3) an introduction to various methods that exploit the RAN platform to facilitate better AI applications, and (4) a discussion of the typical user cases.

The remainder of this paper is organized as follows. Section 2 introduces the hardware architecture of AI-RAN. Section 3 reviews the AI-based approaches for improving RAN performances. Section 4 surveys different AI functions running on the RAN platform. Section 5 provides several use cases, while Section 6 outlines future directions. Finally, Section 7 concludes the paper.

2. AI and RAN

The physical components central to the functionality of AI-RAN are hardware, upgraded from existing RAN infrastructure by integrating high-performance computing resources (typically general-purpose, software-defined hardware accelerators such as GPUs) into access points and/or base stations. Consequently, this infrastructure is expected to achieve a dual goal: (1) to enable data-driven intelligence for conventional RAN services (i.e., AI for RAN) and (2) to serve as a native platform for emerging vertical AI-enabled services (i.e., AI on RAN). As AI and RAN workloads share computational resources at the infrastructure level, this feature is referred to as AI and RAN, as illustrated in Fig. 2. In the following of this section, we detail the structuring element, orchestration

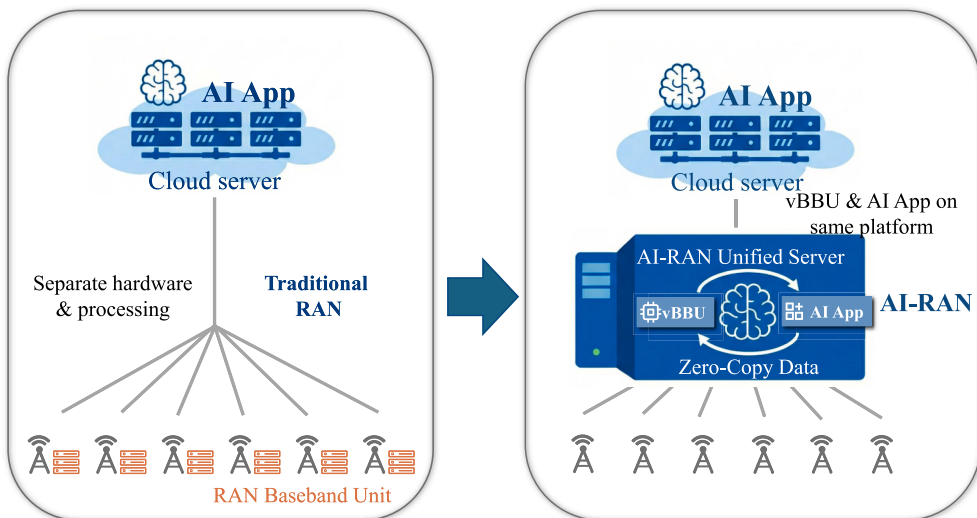


Fig. 1. Illustration of a potential AI-RAN architecture, compared to the traditional RAN.

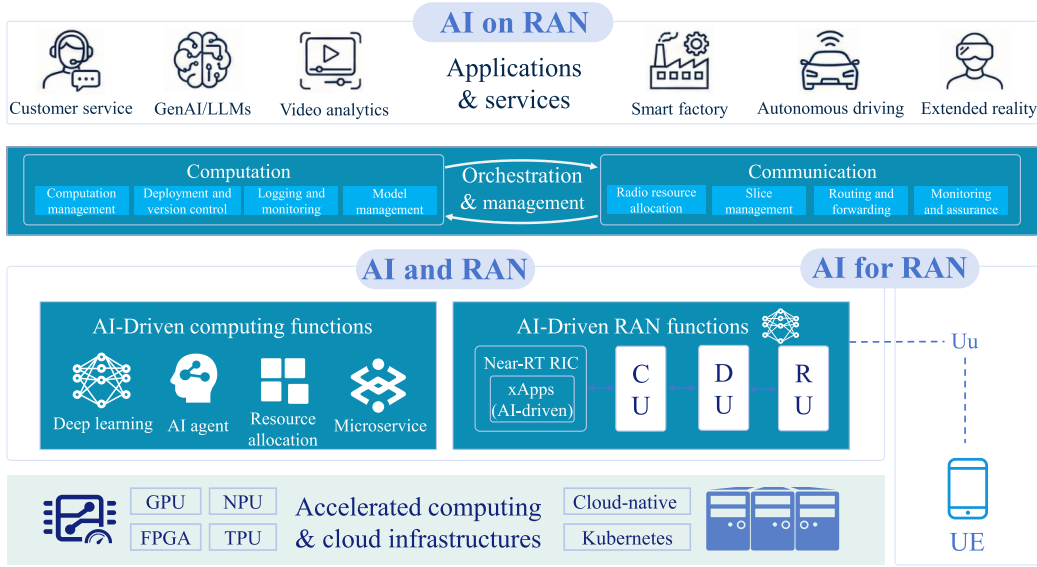


Fig. 2. An illustration of the hardware architecture that enables AI-RAN.

among different workloads and resources, as well as illustrating a demonstrative example of AI-RAN architecture on an O-RAN platform.

2.1. Container-based workload structures

The first step toward seamlessly supporting both AI and RAN workloads on a shared infrastructure is to unify the structuring elements. In the context of AI-RAN, the basic elements constituting the workloads are generally containers. Specifically, the de facto standard for containers is Kubernetes, where the essential components include:

- Pods: The elementary deployment units in Kubernetes. Each pod can encapsulate one or more containers.
- Services: The service abstracts the underlying pods and provides stable endpoints for communication.
- ConfigMaps: This feature manages configuration data and sensitive information for applications.

Correspondingly, we (broadly) categorize the workloads into three groups: The infrastructure management workloads, the platform management workloads, and the application workloads. Specifically, the infrastructure management workloads comprise pods and services that provide infrastructure functionality, including the management of physical resources, networking, and storage. The platform management workloads consist of those pods (and services) responsible for managing platform services, including configuration, orchestration, performance, and security systems. The application workloads, as the name suggests, are the pods and services that execute the AI and/or RAN application workloads.

By encapsulating AI and RAN workloads in containers, we can standardize application formats and specify their dependencies. As a result, container-based workloads can be supported and can be deployed across various platforms with minimal modifications. It thus enables the deployment of applications at scale.

2.2. Orchestration of computational resources for AI and RAN workloads

Computational resources orchestration plays a central role in enabling AI-RAN to simultaneously support AI for RAN, AI on RAN, and ultimately AI and RAN. At its core, this problem concerns how heterogeneous computing resources, i.e., GPU, FPGA, and AI accelerators, are allocated and coordinated to serve both latency-critical RAN baseband processing and computation-intensive AI workloads under a shared infrastructure.

2.2.1. Existing solutions

In current deployments, computation orchestration for AI and RAN is predominantly realized through physical isolation. Specifically, AI and RAN workloads are supported either by time-based or space-based resource partitioning. For instance, [16] presented an AI-RAN proof-of-concept platform based on the multi-instance GPU (MIG) technique, where a fixed fraction of one GPU's resources is dedicated to RAN processing, while the remaining resources are allocated to AI workloads. A similar approach was adopted in

Ref. [17], where CPU resources are statically partitioned in a laboratory testbed to demonstrate the fusion of AI and RAN: a subset of vCPUs is pinned to RAN workloads, whereas the remaining vCPUs are reserved for AI workloads.

Such isolation-based designs simplify timing guarantees for RT RAN functions and avoid mutual interference between communication and AI workloads. However, this approach inherently limits resource utilization efficiency and flexibility. Particularly, idle computing capacity in one domain cannot be easily reclaimed by the other, and cross-layer optimization between communication and intelligence remains largely unavailable. As a result, isolation-based solution is adequate for early-stage AI-RAN deployments, but becomes increasingly inefficient as AI workloads proliferate at the RAN side.

2.2.2. Potential solutions

To overcome the rigidity of isolated designs, AI-RAN advocates a shift toward microservice-based computational orchestration [18]. In this paradigm, both RAN functions (e.g., baseband processing stacks) and AI functions (e.g., training modules, inference engines, feature extractors) are decomposed into fine-grained, containerized microservices. These microservices can be flexibly placed, migrated, and scaled across heterogeneous computing infrastructures under a unified container orchestration framework. By abstracting RAN and AI workloads into a common service layer, microservice-based orchestration enables dynamic sharing of accelerated hardware, elastic scaling according to traffic dynamics, and coordinated scheduling across communication and computation domains [19].

More importantly, microservice-based designs create the foundation for joint orchestration rather than simple co-existence. For instance, AI inference microservices can be co-scheduled with RAN scheduling or beam management functions to minimize control latency, while training or model-update microservices can be deferred to periods of low radio load. Such coordination is difficult to achieve under rigid isolation but becomes natural with containerized microservice orchestration.

2.3. A demonstrative example of AI-RAN architecture on the O-RAN platform

To understand how AI-RAN supports both AI and RAN workloads, we view the AI-RAN architecture through a unified sensing-learning-decision-actuation pipeline, mapped onto RAN functional splits and intelligent control entities. In this pipeline, sensing corresponds to standardized telemetry and data exposure from Radio Unit (RU)/Distributed Unit (DU)/Centralized Unit (CU) as well as from SMO planes; learning encompasses on-node training for fast adaptation as well as cross-node aggregation for global optimization; decision refers to inference and policy instantiation for both RAN configuration and AI service execution; and actuation realizes these decisions through control interfaces that directly affect radio operations or AI task execution. This pipeline is natively supported by the RIC defined in O-RAN, enabling AI-RAN to simultaneously support RAN and AI workloads on shared infrastructures.

As core components of AI-RAN, the near-RT RIC and the non-RT RIC jointly implement a multi-timescale intelligence loop, as illustrated in Fig. 3(a). The near-RT RIC hosts xApps that perform lightweight, inference-driven control over sub-second timescales (10 ms–1 s), interacting with O-DU and O-CU via the E2 interface to regulate scheduling, interference coordination, mobility management, and slice enforcement, thereby supporting latency-sensitive RAN services and AI services. In contrast, the non-RT RIC operates within the SMO domain on slower timescales (≥ 1 s) and hosts rApps for long-horizon policy generation, model training/retraining, and computation-intensive inference for delay-tolerant AI services. Through the A1 and O1/O2 interfaces, it disseminates

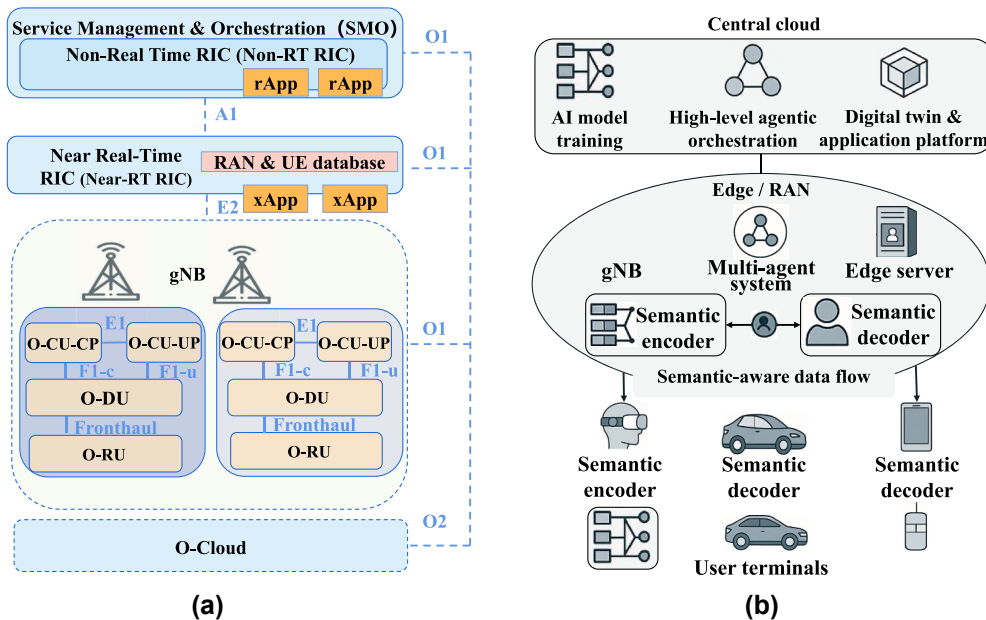


Fig. 3. Illustration of hierarchical RIC and computing in AI-RAN (a) Hierarchical RIC. (b) Hierarchical computing.

learned models and control policies to the near-RT RIC while managing configuration and telemetry. Emerging designs further extend this architecture into the RT domain via node-resident decentralized applications (dApps) deployed directly on RAN nodes, supporting ultra-low-latency control loops or privacy-sensitive AI inference when data export is infeasible. Together, RIC-based xApps/rApps and node-level dApps form a layered AI-RAN architecture that enables AI for RAN intelligence and AI on RAN service provisioning to coexist on shared infrastructure.

The hierarchical RIC architecture naturally gives rise to a cloud-edge-end compute continuum in AI-RAN, as shown in Fig. 3(b). Specifically, RT inference tasks, such as scheduling, mobility management, and semantic processing, are typically executed at DU/CU-adjacent edge nodes or within the near-RT RIC to meet stringent latency requirements, whereas broader model training, aggregation, and latency-tolerant inference are more suitably hosted in regional or central clouds under the non-RT RIC/SMO. In addition, cloud-edge-end collaboration, where each tier contributes complementary capabilities [20,21], is powerful for fast computing and intelligence feedback to enhance the user experience. Collaboration must also be resource-aware, which further motivates a task-oriented communication among cloud, edge, and end-users for communication overhead reduction. Therefore, the AI-RAN architecture is a systematic embedding of the learning loop into a hierarchical control and computing enabled RAN.

3. AI for RAN

AI for RAN represents the application of AI to directly optimize the performance, efficiency, and operation of the RAN. The principal aim is to transition network infrastructure from a static, rule-based model to a paradigm that demonstrates greater adaptability, intelligence, and eventual autonomy. This shift is essential to accommodate the evolving wireless networks, which must address the rapidly increasing and increasingly complex demands characteristic of massive and ubiquitous connectivity. Through AI, RAN can achieve intelligent radio resource management, dynamic traffic steering, predictive load balancing, context-aware energy savings, and so on. Furthermore, it enables advanced capabilities such as automated anomaly detection, root cause analysis, and self-healing mechanisms, significantly enhancing network reliability and service availability.

As shown in Fig. 4, this part provides a structured analysis of how AI augments RAN across multiple layers, i.e., PHY, Medium Access Control (MAC), and network layers, as well as upper layers, enabling enhanced efficiency, adaptability, and intelligence. We examine key AI techniques such as Deep Learning (DL) and RL, and their applications in spectrum management, interference mitigation, resource allocation, and network automation. Challenges including computational complexity, data dependency, and standardization are also discussed. The analysis concludes that AI-driven RAN will be critical to meeting the stringent performance requirements of 5G-Advanced and 6G networks.

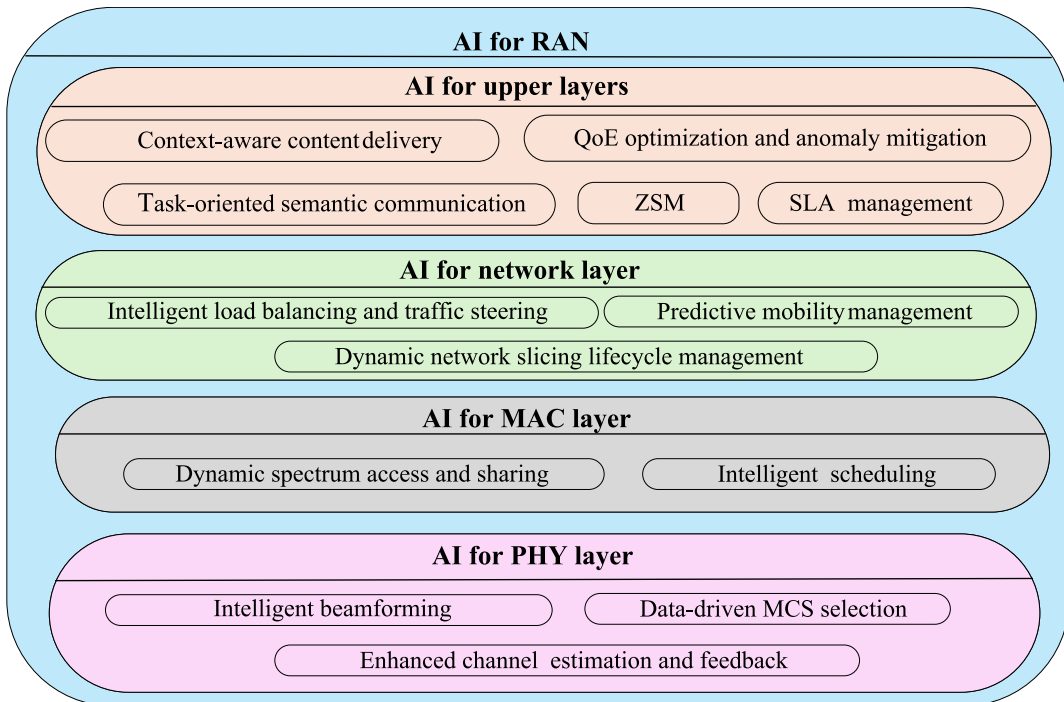


Fig. 4. Layered Application Framework of AI for RAN.

3.1. AI for PHY layer

Conventional PHY algorithms rely on mathematically tractable models that often incorporate simplifying assumptions about channel characteristics, interference, and hardware impairments. AI techniques, particularly DL, circumvent these limitations by learning directly from measurement data, capturing complex, non-linear relationships and hidden patterns that are difficult to model analytically. This enables more robust and adaptive approaches to core PHY functions in dynamic and uncertain radio environments. Some key applications are given as follows.

3.1.1. Enhanced channel estimation and feedback

AI can revolutionize Channel Estimation (CE) and Channel Feedback (CF) in 5G/6G systems, addressing limitations of traditional methods such as high pilot overhead and poor adaptability to complex scenarios [22]. DL excels at capturing spatial-temporal-frequency channel correlations, enabling CE and/or CF optimization for massive Multiple-Input Multiple-Output (MIMO), millimeter-Wave (mmWave), and Reconfigurable Intelligent Surface (RIS)-assisted systems, with frameworks like CEFnet and PFnet demonstrating superior performance [23]. Complementary to data-driven designs, model-driven architectures such as MMV-LAMP integrate physical priors (e.g., angle-delay sparsity) to enhance wideband mmWave CE/CF efficiency [24], while CsiNet pioneers autoencoder-based Channel State Information (CSI) compression that outperforms conventional compression sensing-based methods [25]. For beamspace mmWave systems, DL-based LDAMP networks leverage denoising Convolutional Neural Networks (CNNs) to achieve high accuracy even with limited radio frequency (RF) chains [26]. These AI-enabled solutions consistently deliver lower Mean-Square Error (MSE) and higher spectral efficiency than Least Square (LS)/Minimum Mean Square Error (MMSE) methods, robust to low Signal-to-Noise Ratio (SNR), quantization errors, and dynamic channel conditions, prioritizing adaptive multi-SNR learning and sparsity exploitation to meet future high-mobility and large-scale communication demands.

3.1.2. Intelligent beamforming

Optimizing beamforming vectors in massive MIMO systems is computationally intensive and conventional iterative algorithms may lead to outdated results [27]. Data-driven DL methods shed light on the issue by transferring the online computational complexity to the offline. Hybrid data- and model-driven DL methods, which integrate expert knowledge into neural networks, can further reduce the dependence on the training samples [28]. Leveraging datasets like DeepMIMO constructed from accurate ray-tracing to capture environment-dependent channel characteristics, DL-driven beamforming achieves near-optimal spectral efficiency with negligible inference latency [29]. For dynamic scenarios, meta-learning (e.g., model-agnostic meta-learning) and transfer learning enable fast adaptation to new channel distributions with limited labeled data, outperforming conventional joint training by transferring prior knowledge [30]. Deep unfolding techniques, such as Iterative Algorithm Induced Deep-unfolding Neural Networks (IAIDNN), decompose iterative beamforming algorithms into layer-wise trainable structures, reducing complexity while retaining performance [31,32]. Additionally, [33] proposed a Graph Neural Network (GNN)-based Continual Learning (CL) framework that combines graph-based optimization with adaptive sample selection, to overcome the “catastrophic forgetting phenomenon” in dynamic environments. These AI-enabled solutions consistently improve Signal to Interference plus Noise Ratio (SINR) balancing, robustness to mobility, and spectral efficiency across LOS/NLOS and high-mobility scenarios, laying a foundational framework for efficient beamforming in 5G-Advanced and 6G systems.

3.1.3. Data-driven Modulation and Coding Scheme (MCS) selection

The integration of AI-assisted MCS selection has emerged as a fundamental aspect of adaptive link adaptation. Moving beyond static threshold-based lookup tables, these advanced methodologies leveraging ML paradigms can facilitate data-driven MCS decisions that enhance throughput, reliability, and latency across various contexts [34]. RL emerges as a dominant framework for dynamic environments: contextual multi-armed bandits, Q-learning, Deep Q-networks (DQN), and model-based Thompson sampling allow systems to learn optimal MCS policies through interaction with the channel, maximizing long-term performance without prior channel models. For example, Saxena et al. [35] demonstrated that the adaptation of the RL-based link achieves efficient and tuned-free MCS selection in cellular networks, outperforming conventional methods in rapidly varying channels. DL complements RL by capturing complex non-linear relationships between channel metrics (e.g., SNR, Block Error Rate (BLER), coherence time) and MCS levels: Huang et al. [36] proposed a deep-learning-based design for eMBB/URLLC multiplexing in 5G NR via DL-driven MCS adaptation.

Collectively, these AI-driven solutions enhance MCS selection robustness to imperfect CSI, dynamic channel conditions, and diverse service requirements (e.g., URLLC, eMBB), laying the groundwork for efficient link adaptation in 5G-Advanced and 6G systems.

3.2. AI for MAC layer

The MAC layer is responsible for the critical, time-sensitive coordination of shared radio resources among competing users and services. Traditional algorithms, often based on static policies or optimization models with constrained assumptions, struggle to cope with the extreme heterogeneity, ultra-dense deployments, and rapidly varying traffic patterns of modern networks. AI, particularly RL, empowers this layer by enabling autonomous, context-aware, and proactive decision-making. Instead of relying on pre-defined rules, AI agents can learn optimal control policies directly through interaction with the network environment, dynamically balancing multiple, often conflicting objectives such as throughput, latency, fairness, and energy efficiency. This data-driven approach allows for RT adaptation to unforeseen states and complex scenarios that are not explicitly programmed. Key applications are given as follows.

3.2.1. Intelligent scheduling

User scheduling is the process of efficiently selecting a subset of active users from an available pool and allocating communication resources based on user-specific requirements and priorities. With the proliferation of connected devices and increasing demands for high-speed data transmission, traditional scheduling approaches face significant challenges in scalability, RT adaptability, and fairness optimization. Conventional methods struggle with dynamic network conditions, imperfect CSI, and the trade-off between overall system performance and user fairness. Data-driven methods offer promising alternatives. ML techniques, such as K-means clustering, have been used to group users based on CSI similarity, reducing co-channel interference and improving SINR [37]. DL, particularly GNNs, effectively models network topology and user relationships, enabling near-optimal scheduling with reduced computational overhead [38]. Deep unfolding techniques further accelerate optimization-based scheduling by learning adaptive step sizes. RL provides adaptive and model-free solutions for user scheduling in dynamic environments. For example, a simple multi-armed bandit based online learning approach is employed to learn user scheduling policy for age of information optimization [39].

Despite these advances, challenges remain, including balancing system-wide performance with per-user fairness, adapting to varying user counts, and ensuring scalability in ultra-dense networks. Future work should focus on fair scheduling policies, lightweight algorithms for fluctuating user numbers, and multi-agent learning for interference-aware scheduling in dense deployments.

3.2.2. Dynamic spectrum access and sharing

AI models can learn and predict spatial-temporal spectrum occupancy patterns from historical and RT sensing data. This allows for intelligent, opportunistic access to licensed or unshared bands, and facilitates efficient spectrum sharing between networks while rigorously minimizing harmful interference to primary users. Traditional approaches, including game theory, mixed-integer linear programming, and convex optimization often lack adaptability in highly dynamic and large-scale network environments. RL enables agents to learn optimal allocation policies through interaction with the environment, without requiring explicit system models. For instance, DQN have been used for dynamic spectrum access, improving spectrum utilization efficiency [40]. Policy gradient methods and actor-critic frameworks further enhance RT adaptation to varying network conditions [41]. Applications in 5G/6G networks focus on dynamic spectrum sharing between macro and small cells, as well as scalable allocation in ultra-dense scenarios [42]. Key challenges remain, including minimizing response time for RT applications, optimizing energy efficiency, and ensuring fairness and QoS among heterogeneous users. Future research directions involve advancing RL algorithms for non-stationary environments, integrating cross-layer optimization, and developing robust, privacy-preserving allocation mechanisms for next-generation networks.

3.3. AI for network layer

The network layer optimization focuses on the end-to-end connectivity, routing, and coordinated management of multiple network elements across a wider geographical and temporal scope than individual links or cells. Traditional network optimization often relies on distributed protocols with localized views or centralized planning based on offline models, which can be sub-optimal and slow to react to RT, network-wide dynamics. AI fundamentally empowers this domain by providing a holistic, predictive, and adaptive intelligence plane. By aggregating and analyzing vast, multi-dimensional data streams (e.g., from user equipment, base stations, and the Core Network (CN)), AI models can infer global network states, predict emerging bottlenecks or mobility patterns, and compute coordinated control actions that balance efficiency, resilience, and Quality of Experience (QoE) across the entire RAN. This enables a shift from reactive, local corrections to proactive, global orchestration. Key applications are given as follows.

3.3.1. Predictive mobility management

AI models can analyze historical and RT measurement reports to predict user trajectory, sojourn time, and handover failure risks with high accuracy [43]. This enables proactive and context-aware handover decisions, optimizing parameters for specific scenarios (e.g., high-speed rail, dense urban cells), thereby minimizing ping-pong effects, radio link failures, and signaling overhead in ultra-dense networks. [44] proposes a Privacy-preserving Mobility prediction Framework (PMF) utilizing Federated Learning (FL) to achieve mobility prediction without compromising user privacy, a common issue with centralized models.

3.3.2. Intelligent load balancing and traffic steering

AI can forecast traffic demand hotspots and user distribution across different frequency spectrum and radio access technologies (e.g., 5G NR, LTE). Based on these predictions, it can dynamically adjust cell reselection parameters, handover biases, or steering policies to optimally distribute load, alleviate congestion, and maximize the utilization of heterogeneous network resources. AI can play a pivotal role in addressing the escalating complexity and volume of 5G network traffic—projected to account for 75% of mobile data traffic by 2029 [45]—by enabling accurate traffic prediction, RT optimization, and robust adaptability to dynamic patterns. Traditional time-series models such as Auto-Regressive Integrated Moving Average (ARIMA) and seasonal ARIMA models, while effective for capturing temporal dependencies, struggle with the non-linear, heterogeneous traffic characteristics of 5G networks. In contrast, ML techniques including Support Vector Regression (SVR) and random forests overcome these limitations by modeling non-linear relationships and enhancing prediction robustness through ensemble learning. DL further revolutionizes traffic management. For example, Long Short-Term Memory (LSTM) networks excel at capturing long-term sequential dependencies for traffic prediction [46], transformer-based architectures leverage self-attention mechanisms to process large-scale data efficiently [47], and RL frameworks—such as multi-armed bandit models [48], Deep Deterministic Policy Gradient (DDPG) [49], and decentralized cooperative scheduling approaches [50]—enable adaptive traffic routing, spectrum access optimization, and low-latency scheduling for delay-sensitive and delay-tolerant traffic. Collectively, these AI-driven approaches empower proactive traffic engineering, interference

mitigation, and congestion control, ensuring efficient bandwidth utilization and network resilience amid the exponential growth of 5G traffic.

3.3.3. *Dynamic network slicing lifecycle management*

From instantiation to runtime, ML models predict slice-specific resource demands, perform admission control, and enable elastic scaling of virtualized Network Functions (NFs) [51]. AI-driven continuous monitoring and Key Performance Indicator (KPI) analysis further ensure strict Service Level Agreement (SLA) compliance and effective isolation between concurrent slices sharing the same physical infrastructure. AI has emerged as an indispensable enabler for 6G network slicing, addressing complex challenges in dynamic service provisioning, multi-domain coordination, and ubiquitous network assurance that exceed the capabilities of traditional model-driven approaches. As a core 6G feature, network slicing requires tailored resource allocation, strict service isolation, and seamless global coverage—facilitated by the integration of non-terrestrial components such as Unmanned Aerial Vehicles (UAVs) and satellites. AI techniques have been widely validated to enhance its intelligence and adaptability. For example, in the RAN, RL enables adaptive decision-making for device association and resource allocation in slicing, while FL preserves privacy in distributed scenarios by aggregating local model updates without sharing raw data. In the Transport Network, RL-based elastic bandwidth scaling schemes outperform traditional shortest-path methods in request satisfaction rates, even under dynamic traffic and varying spectrum conditions. In the CN, AI facilitates proactive resource reconfiguration and intelligent anomaly detection through techniques like transfer learning, ensuring reliable operation amid evolving faults and attacks. Beyond individual domains, AI enables automated slice lifecycle management and multi-dimensional SLA decomposition. It leverages supervised and unsupervised learning to map business requirements to network resources across end-to-end slices. Collectively, these AI-driven advancements unlock the six key characteristics of 6G network slicing: tailored service provisioning, efficient resource utilization, strict isolation, seamless coverage, technology convergence, and ubiquitous intelligence. This foundation supports differentiated services in vertical applications such as holographic communication and autonomous driving.

3.4. *AI for upper layers*

Traditionally, the upper layers have operated independently from the underlying network, with applications adapting to network conditions in a reactive, best-effort manner. AI disrupts this paradigm by establishing a dynamic, bidirectional intelligence loop between the applications and the RAN. By exposing RT network capabilities, predicted states, and contextual awareness to applications—and conversely, by ingesting application intent, content characteristics, and user behavior—AI enables the network to become service-aware and applications to become network-aware. This symbiotic intelligence facilitates proactive resource provisioning, personalized service delivery, and the creation of novel, immersive experiences that are tightly coupled with network performance. Some key applications are given as follows.

3.4.1. *SLA management*

AI models can translate high-level application requirements into precise, dynamic network-level policies. By continuously monitoring both application KPIs (e.g., video buffer health, gaming frame jitter) and network conditions, AI can predict SLA violations and trigger pre-emptive resource allocation or path reconfiguration for premium users or critical sessions, ensuring a consistently high QoE [52].

3.4.2. *Context-aware content delivery*

By analyzing user behavior patterns, content popularity trends, and local context, AI can predictively store relevant data and deploy necessary compute functions (e.g., video transcoding, AI inference models) at optimal edge locations [53]. This minimizes latency, reduces backhaul load, and enables RT Augmented Reality (AR), Virtual Reality (VR), and interactive services.

3.4.3. *Predictive QoE optimization and anomaly mitigation*

Moving beyond network-centric KPIs, AI correlates multi-domain data—including user device metrics, application logs, and network performance—to model and predict the subjective QoE. It can identify the root cause of user-perceived degradation (e.g., a video stutter caused by a radio issue versus a server-side problem) and orchestrate targeted mitigation actions, such as switching a video stream to a more robust codec or adjusting scheduling priority. Unsupervised learning models identify anomalies in network KPIs, facilitating rapid fault detection and root cause analysis [54].

3.4.4. *Zero-touch network and Service Management (ZSM)*

Traditional RAN O&M relies heavily on manual intervention, which is inefficient and error-prone, especially in large-scale and complex networks. AI enables ZSM by automating intent interpretation, predictive analysis, network configuration, and service provisioning [55]. ML models trained on historical network data can predict potential faults (e.g., base station failures, link degradation) and trigger proactive maintenance, reducing downtime and operational costs.

3.4.5. Task-oriented semantic communication

Task-oriented semantic communication, empowered by AI, shifts the core objective from signal reconstruction to task accomplishment. Under this paradigm, the key performance indicator becomes the probability of task success, such as classification accuracy, control stability, or detection miss rate, within given communication constraints. Semantic efficiency, in turn, measures how many task-relevant units can be transmitted per unit spectrum or energy [13,15]. Recent studies highlight that this transition is far from superficial: once task utility replaces bit fidelity as the benchmark, the optimal operating point across the rate–latency–reliability space fundamentally shifts. This enables cross-layer designs that jointly leverage communication and intelligence, making integrated optimization feasible. Therefore, for AI-RAN, semantics serves as a new guiding principle for air-interface evolution—particularly when the RAN must function as the foundational layer for native AI services.

4. AI on RAN

Beyond using AI to improve network performance, conversely, the RAN infrastructure can also be utilized as a supporting platform to facilitate AI applications and services, which is referred to as AI on RAN. Recent studies have increasingly explored the utilization of RAN infrastructure as a platform for supporting AI-driven applications, particularly in vertical domains such as smart cities and GenAI/LLMs [56,57]. This line of research highlights an emerging convergence between RAN and AI within the telecommunications ecosystem. From a broader perspective, AI on RAN is widely regarded as a promising paradigm that can enhance the value of existing RAN resources by enabling AI as a Service (AIaaS), serving both internal operational functions (e.g., network management, customer support, marketing analytics, and fraud detection) and external enterprise-oriented use cases.

In the construction of 6G-oriented AI on RAN architecture, edge computing has evolved from an optional value-added service into the fundamental enabling substrate of the intelligence-native network. AI on RAN not only deploys AI models at the network edge but also integrates edge computing capabilities, transforming the RAN from a traditional connectivity pipeline into a native intelligent platform that embodies the trinity of communication, computation, and intelligence. This integration not only supports low-latency, high-reliability RT inference but also provides the infrastructure for distributed collaborative learning—thereby ensuring both the autonomous efficiency of the network itself and enabling intelligent applications across vertical industries. The following sections will systematically elaborate on the technical essence, enabling mechanisms, and synergistic relationships of edge learning and edge inference.

4.1. Edge learning

Edge learning constitutes the distributed intelligence engine of AI on RAN, enabling model training and continuous adaptation directly within the network's access layer without necessitating raw data centralization. This paradigm is fundamentally driven by the intrinsic distribution and locality of RAN data—such as channel state information, user mobility patterns, and slice-specific KPIs—which are generated at geographically dispersed base stations and user devices. Centralized cloud-based training is often impractical due to fronthaul/backhaul constraints, latency sensitivity, and growing data privacy regulations [58,59]. Edge learning addresses these constraints by keeping data at its source and moving the learning process to the network periphery.

The primary technical realization of edge learning in AI on RAN is FL. In this framework, distributed RAN nodes (e.g., O-DUs, near-RT RICs) or even user equipment collaboratively train a shared global model. Each participant performs local training on its native dataset and exchanges only model updates (e.g., gradients or weights) with a central aggregator, often co-located with the non-RT RIC or a regional cloud. This approach preserves data privacy, reduces CN load, and allows models to learn from spatially heterogeneous yet representative local conditions. FL naturally aligns with the O-RAN architecture, enabling, for instance, collaborative policy optimization among xApps across multiple vendors or operators without exposing proprietary data.

To further tailor learning to the RAN's hierarchical and latency-stratified nature, Hierarchical and Split Learning architectures have emerged [60,61]. Hierarchical FL introduces intermediate aggregators at the edge or CU level, mirroring the RAN's DU-CU-edge-cloud topology. This reduces communication overhead to the central coordinator, improves convergence speed, and enhances robustness against straggler nodes. Split learning partitions a model vertically: early layers are executed on devices or DUs close to the data source, while later, computationally intensive layers are processed at the edge or cloud. This not only alleviates the compute burden on constrained devices but also provides an additional layer of data privacy by preventing raw data from leaving the local node.

However, deploying edge learning in AI on RAN presents distinct RAN-specific challenges and opportunities as shown in Fig. 5. The non-IID and skewed nature of local data across cells can severely impede FL convergence and model fairness. A cell in a dense urban area generates data fundamentally different from a rural cell. Furthermore, resource dynamics and straggler effects are exacerbated in wireless environments, where participants may suffer from temporary radio resource scarcity or computational bottlenecks, delaying the global aggregation cycle. Communication efficiency is paramount, as frequent model update exchanges consume valuable fronthaul resources. Techniques like over-the-air computation, gradient compression, and sparse update schemes are essential to make edge learning feasible. Finally, security and trust in a federated ecosystem remain paramount, requiring robust aggregation rules, secure multi-party computation, and mechanisms to detect and mitigate data or model poisoning from malicious or malfunctioning nodes.

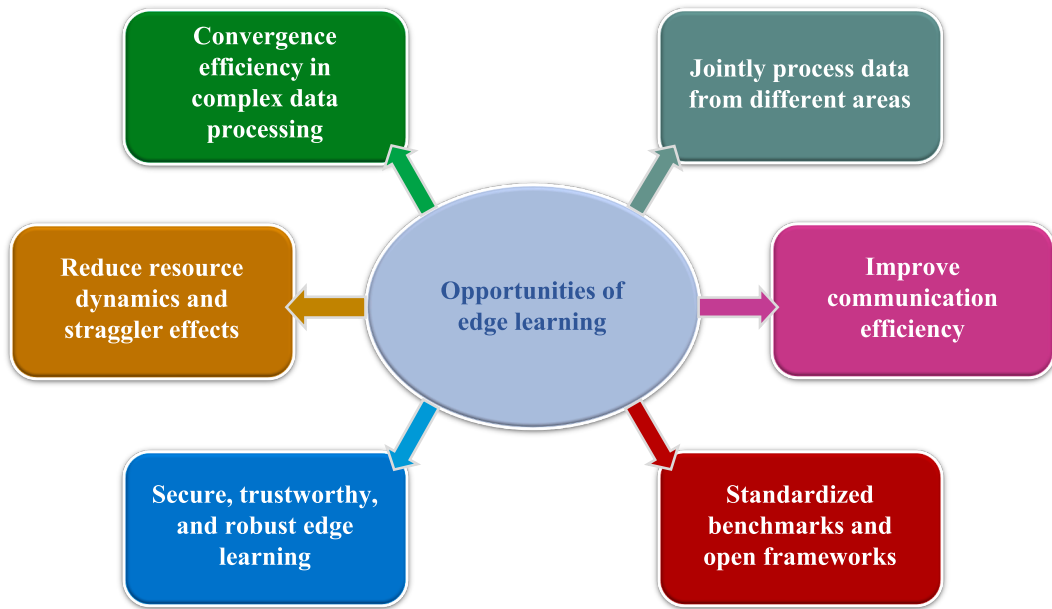


Fig. 5. The opportunities of edge learning.

4.2. Edge inference

While edge learning forms the distributed “cognitive system” of AI on RAN, enabling models to understand and adapt to the network environment through collaborative training, this intelligence remains latent without a corresponding “executive system” for RT actuation. Edge inference translates trained models into immediate decisions that close the control loop. Together, edge learning and edge inference establish a complete intelligent closed-loop, where the network continuously learns from distributed data and instantly applies that learning to optimize its own operation and deliver advanced services. This synergy is fundamental to AI on RAN, moving beyond episodic optimization to achieve embedded, perpetual, and responsive autonomy.

The primary function of edge inference is to deliver low-latency, high-availability AI decision-making directly within the RAN, thereby fulfilling the RT requirements of RAN control and user-facing applications. This capability is a functional necessity, as the physical constraints of propagation delay and the sub-second timing requirements of tasks like scheduling, beamforming, and handover preclude reliance on distant cloud data centers.

Edge inference is fundamentally architected around the principle of proximity. By co-locating AI inference engines with baseband processing units (e.g., on GPU/NPU-accelerated O-DU servers or near-RT RICs), AI on RAN minimizes control-to-action latency. This co-location is exemplified by implementations such as Beamforming Neural Networks (BNNs) [27,28], which can execute inference in milliseconds at the edge to translate channel state into optimal beamforming weights, embodying the fusion of communication and AI computation.

To achieve scalability and efficiency, edge inference relies on sophisticated cloud-edge-device collaboration and model splitting. A split inference pipeline is employed, where lightweight initial model layers run on devices or far-edge nodes, and deeper layers execute on more powerful near-edge or cloud infrastructure. This strategy aligns with the O-RAN functional decomposition, conserving resources and balancing computational load across the network hierarchy.

However, as shown in Fig. 6, deploying inference at the edge introduces a complex set of system trade-offs that must be jointly optimized, primarily the accuracy-latency-energy trilemma. Resource contention on shared servers between baseband and AI tasks can also lead to non-deterministic latency. Therefore, joint radio-compute orchestration is critical, applying principles from multi-access edge computing research to co-schedule radio and compute resources for superior performance. This requires edge inference optimization mechanisms such as predictive model caching and pre-positioning to proactively load anticipated models and eliminate startup delay; context-aware adaptive inference to dynamically adjust execution parameters based on real-time network load or application priority; and inference-aware mobility management to seamlessly migrate session states during handover, thus ensuring service continuity and efficient resource utilization in AI-on-RAN systems.

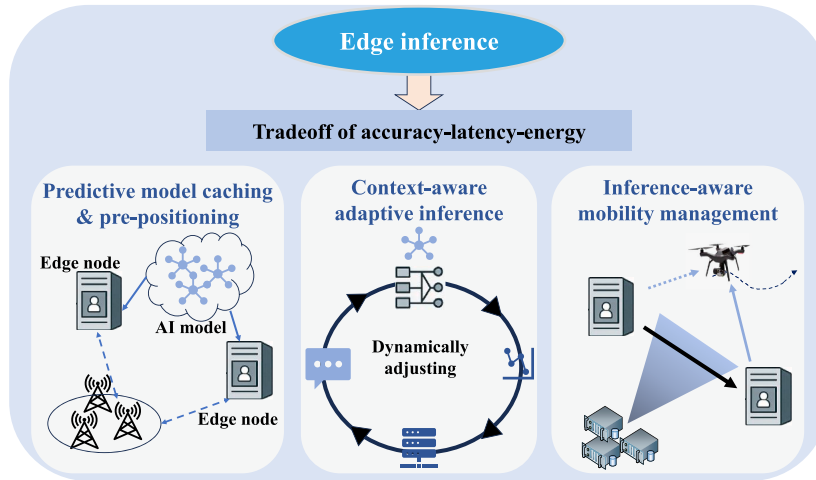


Fig. 6. The opportunities of edge inference.

The proposal of AI on RAN marks a paradigm shift of the RAN from a connectivity pipe to an intelligent platform. Its core objective is to open up RAN computing, storage, and communication resources to support various low-latency, high-reliability, and distributed AI applications while maintaining network performance. This architecture inherently relies on two key technical pillars: edge learning and edge inference. In conclusion, edge inference provides the essential nervous system for RT actuation in AI on RAN. When integrated with the learning brain of edge learning, they form a complete closed-loop intelligent system that allows the RAN to perceive, understand, and adapt in RT, fulfilling the promise of a truly autonomous 6G network.

5. Case study

AI-RAN has emerged as a pragmatic approach to enhance RANs by embedding data-driven intelligence into optimization and control loops while respecting carrier-grade constraints. Typical AI-RAN use cases include: traffic steering and load balancing across multi-layer cells, energy saving via cell/beam sleep and power adaptation, mobility robustness optimization and handover parameter tuning, beam management and CSI acquisition reduction for massive MIMO, interference coordination in dense deployments, and anomaly detection and self-healing for operational resilience. This section presents three representative case studies, selected for their deployability and their complementary timescales—from sub-10 ms beam decisions to minutes-level energy optimization—highlighting topology, implementation details, and end-to-end gains.

In the following, we present two representative use cases in detail: (1) Energy Saving rApp (ES-rApp) energy saving exemplifies AI for RAN, as AI-driven predictions are translated into bounded radio configuration actions to optimize RAN energy-QoE tradeoffs; (2) INA-Infra intent-driven slicing is primarily AI on RAN, emphasizing end-to-end intent translation, cross-domain orchestration and SLA assurance, while leveraging RAN control as a downstream actuator.

5.1. AI-native energy saving via cell sleep with QoE guarantees

Energy consumption has become a first-order concern in modern RAN operations, particularly for private/enterprise deployments where traffic demand is highly non-stationary across time and space. Static power-reduction policies (e.g., fixed offsets or periodic sleep) are typically conservative (e.g., leaving savings on the table) or aggressive (e.g., causing QoE degradation when demand surges). This case study presents an AI-driven ES-rApp built on a practical AI-RAN control loop consistent with a Non-RT RIC rApp architecture, which includes telemetry-driven intelligence, standardized O1 actuation, and explicit verification to ensure carrier-grade reliability [62].

As shown in Fig. 7, we consider a two-cell enterprise topology representative of real deployments: an indoor small cell and an outdoor cell operating in the same band with partially overlapping coverage. The outdoor site includes multiple RUs serving corridors and peripheral zones that frequently become idle. A Non-RT RIC hosts rApps and interfaces with the SMO, while the RAN exposes telemetry and configuration knobs via standardized interfaces. This setting supports hierarchical control: Non-RT intelligence computes policies on minute-level timescales, while the RAN executes fast fallback behavior locally when needed.

ES-rApp follows a predict-decide-verify pipeline designed for operational safety and explainability. To be specific, (1) short-horizon demand prediction: ES-rApp consumes time-windowed KPM (Key Performance Measurement) sequences (e.g., PRB (Physical Resource Block) utilization, throughput, MCS distribution, RSRP (Reference Signal Received Power)/RSRQ (Reference Signal Received Quality)/SINR trends) to forecast near-term load and service demand. Temporal models (lightweight DNN/LSTM variants) are preferred because real RAN data often exhibits strong autocorrelation; moreover, your prior deployment demonstrates that sequence-based

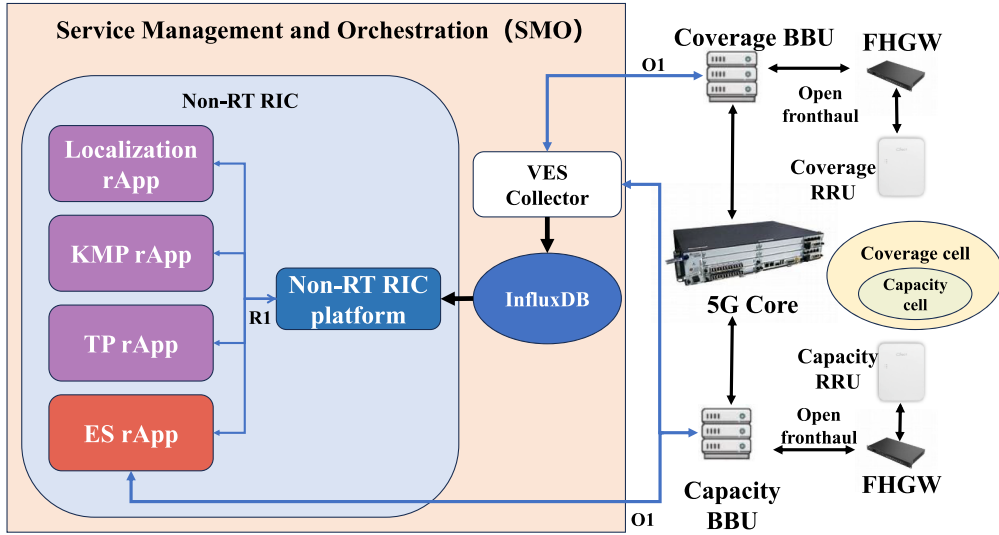


Fig. 7. System architecture of AI-driven energy saving application (ES rApp) in real-world 5G deployment, where TP rApp stands for the Traffic Prediction application, KMP rApp stands for Key Performance Measurement collection and analysis application.

predictors can achieve high accuracy with low inference latency suitable for online decision loops. (2) Discrete energy-state selection with guardrails: To avoid unstable behavior and simplify validation, ES-rApp selects among discrete energy states per RU/carrier: normal, mild saving (−3 dB RU TX power), and deep saving (−5 dB RU TX power, optionally with carrier/beam muting when footprint is predicted idle). This actuation pattern intentionally mirrors an engineering-proven configuration style, where RU TX power is adjusted through O1 interface with 3 dB/5 dB steps, which is interpretable and easy to audit. To protect QoE, ES-rApp enforces guardrails including minimum coverage/SINR thresholds, demand-triggered wake-up constraints, bounded update rates, and KPI-based alarms that trigger automatic rollback. (3) Closed-loop verification: A PM-style verification stage confirms that intended configurations take effect, preventing silent failures and enabling robust troubleshooting.

ES-rApp achieves savings primarily through high-duty-cycle power scaling in spatially idle footprints and off-peak periods. Because the design uses standardized 3 dB and 5 dB steps, each action provides a meaningful reduction in transmit power (and hence energy) when applied persistently under low demand. Recommended evaluation metrics include energy saving ratio, coverage robustness, user throughput/latency, handover stability, rollback rate, and actuation success rate and latency. Operationally, the key advantage is that ES-rApp embodies a verifiable and auditable AI-RAN loop: prediction drives discrete, bounded actions; O1 enforces carrier-grade configuration; PM confirms effect and supports safe rollback. Remaining challenges include handling bursty traffic, multi-cell coupling, and lifecycle robustness. Overall, this case study illustrates a pragmatic pathway to sustainable AI-RAN that is deployable, interoperable, and reliability-aware.

5.2. Intent-driven and AI-assisted network automation for AI-RAN

As RANs evolve to support heterogeneous services, including eMBB, URLLC, and mMTC, with rapidly changing QoS, automation must extend beyond radio scheduling to include compute placement and X-haul coordination from edge to cloud. INA-Infra provides an open and extensible infrastructure that operationalizes this vision through intent-driven network automation: operators specify high-level intents (e.g., latency/throughput targets per traffic class), while the system automatically translates them into low-level actions across the RAN and core, including end-to-end network slicing and resource orchestration [63].

As shown in Fig. 8, an Intent-driven Network Automation infrastructure (INA-Infra) is developed. INA-Infra is built on an O-RAN-aligned, cloud-native architecture spanning three distributed data-center pools—edge, regional, and central—each hosting a Nephio workload cluster and containerized NFs. The testbed integrates (1) SMO + O-Cloud (IMS/DMS) for lifecycle and infrastructure management via O2/O1, (2) non-RT RIC for intent-based network management and rApp orchestration, and (3) near-RT RIC (FlexRIC) for RT control via E2 using xApps. The RAN is implemented with OAI CU/DU/RU, and the core uses OAI 5 GC with slicing support (NSSF (Network Slice Selection Function)/AMF (Access and Mobility Management Function)/SMF (Session Management Function)/UPF (User Plane Function)). This topology enables hierarchical automation: the non-RT layer translates intents and coordinates multi-domain resources, while the near-RT layer enforces slice-level radio actions (e.g., PRB allocation) on short timescales.

In multi-slice networks, meeting slice SLAs depends on both where slice-specific NFs (e.g., CU-UP/UPF/SMF) are deployed and how radio resources are allocated. Deploying NFs closer to the edge reduces latency but incurs higher resource cost; central deployment is cheaper but may violate URLLC-like delay budgets. INA-Infra targets this joint optimization by enabling the operator to specify intents (delay/throughput per slice), while the platform automatically performs placement, scaling, and radio slicing actions to satisfy SLAs.

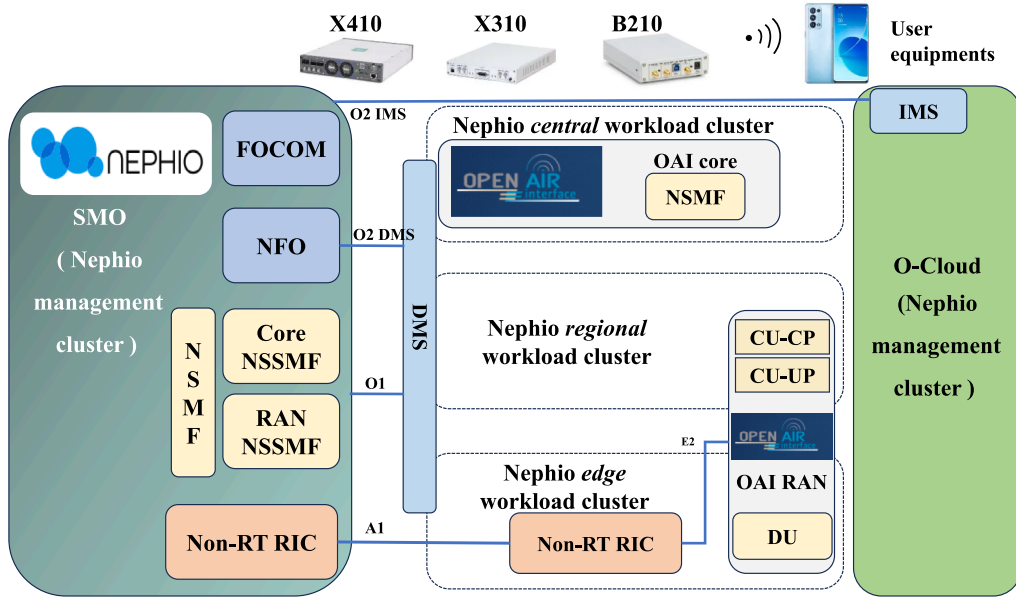


Fig. 8. System architecture of INA-Infra testbed in real-world open 5G network, where FOCOM stands for SMO's Federated O-Cloud Orchestration and Management module, DMS is short for Deployment Management Services, Nephio is a Kubernetes-based project aimed at enabling intent-driven automation for network functions and infrastructure, NSMF stands for Network Slice Management Function, Core-NSSMF stands for Core Network Slice Subnet Management Function, RAN-NSSMF stands for the RAN Network Slice Subnet Management Function, and CU-CP and CU-UP are short for control plane and user plane in centralized unit, respectively.

INA-Infra combines Nephio's declarative intent automation with O-RAN intelligent control. Step 1: intent specification. The operator defines slice-level SLA targets (delay and throughput) at the SMO/NSMF (Network Slice Management Function). Step 2: NF placement and lifecycle automation. Nephio orchestrates containerized CU-UP (Central Unit - User Plane)/UPF (User Plane Function)/SMF (Session Management Function) deployments across edge/regional/central clusters, implementing the requested slice architecture (shared control plane; per-slice user plane). Step 3: closed-loop assurance. RM-rApp monitors slice KPIs via O1; if SLA is violated, it scales or reallocates compute resources and issues A1 policies. Step 4: near-RT enforcement. NS-xApp executes E2 actions to adjust slice PRB shares in the DU, closing the loop on radio resource control. Step 5: KPI validation. Slice throughput and RTT (Round Trip Time) are measured (e.g., iperf3 and ping) to verify SLA satisfaction under multi-slice load.

Experiments with four slices show that as additional User Equipments (UEs)/slices are activated, the system maintains slice throughput targets through PRB adaptation, while RTT measurements meet delay SLAs (evaluated under controlled traffic conditions). Moreover, a key empirical insight is that slice throughput degrades when CU-UP CPU quotas are reduced, revealing a near-linear coupling between compute allocation and user throughput. This validates the need for an automation loop that treats compute as a first-class resource, not merely a deployment artifact. Under compute overload at the Edge cluster, introducing RM-rApp reduces SLA violation for the highest-priority slice, demonstrating the practicality of intent-driven prioritization in real testbeds. This case study highlights a core AI-RAN principle: Automation must be end-to-end and multi-domain, integrating radio, compute, and orchestration across distributed infrastructure. Overall, the INA-Infra use case provides a reproducible blueprint for AI-RAN experimentation where intelligence is not isolated to a single controller, but embedded into a hierarchical, intent-driven automation stack spanning edge-to-cloud.

6. Future research directions

In this section, we identify several future research directions.

6.1. Software and hardware co-design

Realizing AI-RAN at carrier scale requires moving beyond deploying AI on RAN data toward joint optimization of learning pipelines, protocol functions, and heterogeneous compute across RU/DU/CU and edge/cloud. A key future direction is cross-layer co-design in which compute becomes a first-class resource, co-scheduled with time-frequency radio resources. This enables compute-aware RAN control: adaptive feature acquisition, model selection, and inference cadence that respond to channel dynamics and compute headroom, while aligning functional split decisions with AI workload latency and bandwidth demands. A second direction is hardware-aware learning and model-architecture co-optimization. Generic compression (quantization/pruning) is insufficient because model errors can directly degrade radio KPIs or destabilize control loops. Future AI-RAN platforms should adopt KPI-constrained

neural architecture search and multi-objective optimization that minimize tail latency and jitter under strict reliability constraints. This also motivates domain-specialized accelerator primitives (e.g., kernels for beam prediction, graph-based interference modeling, and channel feature extraction) and hybrid precision designs where safety-critical decision modules use higher precision or structured models. Third, AI-RAN demands deterministic execution and jitter-resilient inference. Many RAN loops operate at sub-10 ms time-scales, while AI inference may exhibit variable latency due to memory contention and resource sharing. Research is needed on predictable runtimes and mixed-criticality isolation, complemented by graceful degradation mechanisms (e.g., cached features, multi-rate control, or approximate inference) that preserve stability and QoS under overload. Finally, distributed co-design across network and compute fabrics will be essential. Model partitioning, in-network computing (e.g., DPUs/SmartNICs for aggregation), and accelerator-cooperative training can reduce fronthaul load and enable continual adaptation, but require synchronization and consistency mechanisms that prevent performance drift. To operationalize these advances, AI-RAN must also embed observability (AI-RAN performance counters), lifecycle automation (safe updates/rollback), and energy/carbon-aware scheduling to ensure sustainable deployment. Overall, software–hardware co-design will define whether AI-RAN evolves from prototypes to interoperable, reliable, and scalable network intelligence.

6.2. Protocol design towards Internet of agents and embodied intelligence

To enable the Internet of agents and embodied intelligence, protocol design for AI-RAN infrastructures would extend beyond the traditional (layered) communication architectures that are normally optimized for packet deliveries. Specifically, protocol stacks in conventional wireless networks abstracts away application-related semantics, focusing on reliability, latency, and spectral efficiency under relatively static traffic models. In contrast, AI-RAN is anticipated to support large-scale deployment of agents, which generate traffic tightly coupled with sensing, inference, and control processes. This necessitates protocol primitives that are *task-aware*, enabling the network to differentiate between information critical for RT decision-making and data processing that can be delayed, compressed, or even compromised. As a result, protocol design shifts from flow-based optimization toward *control- and learning-driven* communications, where scheduling, resource allocation, and retransmission policies are dynamically adapted based on agent objectives, uncertainty levels, and closed-loop performance requirements rather than purely channel-centric metrics.

In addition, unlike conventional networks where intelligence resides primarily at the endpoints, AI-RAN protocols actively support intermediate processing such as semantic filtering, model aggregation, and selective state exchange across the base stations. This requires new cross-layer interfaces that expose learning states, model updates, and response deadlines to the network, allowing the RAN to co-optimize communication resources and computational tasks. Moreover, protocols must accommodate heterogeneous agents with diverse sensing rates, actuation constraints, and learning dynamics, while remaining robust to partial observability and non-stationary environments. Consequently, AI-RAN protocol stacks evolve toward *adaptive*, learning-enabled architectures that continuously *reconfigure* themselves based on observed agent behaviors and network conditions, transforming the RAN from a passive transport layer into an active coordination and intelligence-enabling platform.

6.3. Sensing-communication-computation-control closed loop

An important open issue in AI-RAN is the design of a holistic sensing-communication-computation-control closed loop, which is fundamental to achieve scalable, reliable, and autonomous intelligence provisioning in practical deployments. For example, in autonomous driving scenarios, safety-critical control depends on continuously updating learned models and control actions based on newly sensed data, forming a tight closed loop to ensure timely and safe vehicle operation. Similarly, in immersive XR and metaverse services, user motion and environmental sensing must be tightly coupled with wireless transmission, edge rendering, and adaptive control of bitrate, latency, and viewpoint prediction, so that perceptual quality and motion-to-photon responsiveness can be maintained through a closed-loop interaction.

While existing research enables partial integration of these components, achieving a unified closed-loop design remains challenging due to several fundamental reasons. First, sensing data are often noisy, unlabeled, delayed, and spatio-temporally heterogeneous, whereas learning and control modules typically assume clear, accurate, and well-labeled data samples; bridging this mismatch requires principled approaches to data cleaning, uncertainty modeling, and representation/continual learning. Second, although communication and computation resources are tightly coupled in AI-RAN, they are still largely orchestrated independently, making it difficult to guarantee end-to-end latency and system stability when control decisions depend on both radio dynamics and computational workload states. Third, ensuring closed-loop stability and safety under time-varying and partially observable environments further complicates the problem, especially when data distribution and network resources evolve across temporal and spatial dimensions. Addressing these challenges calls for cross-layer theories and system designs that jointly consider sensing accuracy, communication reliability, computational availability, and control robustness, ultimately enabling trustworthy and self-evolving closed-loop intelligence in AI-RAN.

6.4. Self-evolving learning

Self-evolving learning has emerged as a pivotal research direction toward achieving autonomous and continuously optimizing network intelligence in the evolution of AI-RAN. As AI models increasingly handle dynamic decision-making and resource allocation tasks in RAN environments, traditional static training paradigms fall short in addressing the non-stationarity and complexity of

wireless networks. Self-Evolving Learning, which integrates meta-learning, RL, and neural architecture search, enables AI agents to learn how to autonomously generate, evaluate, and improve their own learning structures and strategies during operation.

Recent studies highlight the potential of this paradigm for adaptive and robust network intelligence. For instance, Bhukya et al. [64] proposed the Autonomous Evolutionary Meta-Learning (AEML) framework, demonstrating how AI systems can autonomously create new models without human intervention. Similarly, Tang et al. [65] and Shi et al. [66] extended this concept to networked intelligence with a self-evolving edge learning architecture, allowing edge nodes to adaptively optimize performance under fluctuating communication conditions and continual data arrivals. Moreover, Gao et al. [67] emphasized that self-evolving agents represent a critical step toward Artificial Super Intelligence (ASI), as they continuously evolve their objectives and learning strategies through environmental feedback. Self-evolving learning will provide AI-RAN with the theoretical and technical foundation for building self-organizing, self-healing, and self-optimizing wireless intelligence systems. Through these advancements, AI-RAN systems are expected to evolve from passive learners to proactive, autonomous intelligent networks capable of continuous self-improvement.

Achieving self-evolving learning in AI-RAN needs to overcome several key challenges related to theoretical grounding, stability, efficiency, and interpretability. Current frameworks, such as AEML [64], demonstrate the potential for autonomous model evolution but lack unified mathematical formulations to ensure convergence and controllability. The non-deterministic nature of evolving models in Ref. [67] also raises stability and safety concerns. Furthermore, deploying self-evolving learning in resource-constrained environments like AI-RAN requires lightweight, communication-efficient algorithms [65,66]. Addressing these issues is crucial for developing scalable, trustworthy, and truly autonomous self-evolving AI-RAN systems.

7. Conclusion

The transition from traditional, hardware-centric architectures to the AI-RAN paradigm marks a fundamental shift in telecommunications infrastructure. As demonstrated in this paper, the convergence of AI and RAN is no longer a theoretical pursuit but a practical necessity, driven by the urgent demand for resource efficiency, ultra-low latency, and autonomous network management. Throughout this survey, we have systematically mapped this convergence across three interdependent aspects, namely, AI and RAN (infrastructure integration), AI for RAN (performance enhancement), and AI on RAN (service delivery). By analyzing current progress within these domains, we have identified both the transformative potential and the significant research challenges that remain. Ultimately, AI-RAN stands as the cornerstone of the 6G era. By unifying intelligence and connectivity, this framework provides a definitive roadmap for a more flexible, efficient, and natively intelligent telecommunications ecosystem.

CRedit authorship contribution statement

Chenyuan Feng: Writing – original draft. **Howard H. Yang:** Writing – review & editing. **Kun Guo:** Writing – review & editing. **Wenchao Xia:** Writing – review & editing. **Chenxi Liu:** Writing – review & editing, Conceptualization. **Tony Q.S. Quek:** Conceptualization.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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